

Highlights

Efficient approximation of vessel wall stress under pulsatile hemodynamics using a snapshot-based coupling framework

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- A snapshot-based framework estimates vessel-wall stress under pulsatile flow.
- 1D reduced-order hemodynamics provide time-specific boundary conditions.
- Steady 3D fluid and structural analyses approximate instantaneous FSI stress.
- The framework was validated against fully coupled 3D FSI simulations.
- Computational cost was reduced by over 30-fold across vascular models.

Efficient approximation of vessel wall stress under pulsatile hemodynamics using a snapshot-based coupling framework

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Abstract

Fluid–structure interaction (FSI) analysis is essential for evaluating vessel-wall stress under pulsatile hemodynamic loading, but fully coupled three-dimensional (3D) FSI simulations remain computationally prohibitive for repeated or patient-specific applications. This study presents a snapshot-based computational framework for efficient estimation of localized vessel-wall stress at selected phases of the cardiac cycle. The proposed method combines a one-dimensional (1D) reduced-order hemodynamic model with steady 3D fluid and structural analyses. First, pulsatile pressure and flow-rate waveforms are computed using a physics-based 1D reduced-order model. Boundary conditions are then extracted at representative time instants and applied to steady 3D fluid simulations to reconstruct wall traction. Finally, the transferred traction is used in a steady structural analysis to estimate the corresponding vessel-wall stress. The framework was validated against fully coupled pulsatile 3D FSI simulations in three cases: an idealized left anterior descending coronary artery model, a subject-specific coronary artery model, and a subject-specific abdominal aortic aneurysm model. Across all cases, the proposed framework reproduced the von Mises stress field with low global relative L_2 -norm error at the selected time instants. The error remained below 1% in the idealized coronary model, below 5% in the subject-specific coronary

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model, and below 2% in the aneurysm model. Larger discrepancies were observed during phases of rapid wall motion, whereas smaller errors were found at end-diastole and mid-diastole. The proposed framework also substantially reduced computational cost over 30 folds compared with fully coupled 3D FSI simulations.

Keywords: Fluid–structure interaction, Vessel wall stress, Reduced-order modeling, Pulsatile hemodynamics, Snapshot-based coupling

1. Introduction

Cardiovascular computational modeling has become an essential tool for the quantitative analysis and prediction of complex physiological phenomena [1, 2]. Because cardiovascular systems are inherently nonlinear and dynamically coupled, with interactions spanning flow, pressure, wall deformation, and downstream vascular response, conventional morphology-based assessment alone is often insufficient to characterize the underlying biomechanical environment [3, 4]. In particular, the vascular system is subjected to intrinsically pulsatile hemodynamic loading generated by the cardiac cycle, and the vessel wall deforms continuously in response to this time-varying flow field. Accordingly, hemodynamics and wall mechanics cannot be decoupled without loss of physical fidelity, which has motivated extensive use of computational fluid dynamics (CFD) and fluid–structure interaction (FSI) analyses in cardiovascular biomechanics [5, 6, 7, 8]. This need is also clinically relevant. Current clinical assessment of vascular lesions still relies primarily on anatomical imaging, luminal severity measures, and, where applicable, pressure-derived physiological indices to estimate lesion significance [9, 10, 11]. However, lesions with similar morphological characteristics may exhibit different local flow fields, wall shear stress distributions, and vessel-wall stress states, which are more directly associated with disease progression and rupture risk [12, 13, 14, 15, 4]. CFD enables detailed characterization of velocity, pressure, and wall shear stress, whereas coupled fluid–structure analysis further provides access to wall deformation and intramural stress under physiologically realistic loading conditions [16, 8].

Despite these advantages, high-resolution three-dimensional (3D) pulsatile simulations, particularly fully coupled FSI analyses, remain computationally demanding and are therefore difficult to apply in repeated studies, large patient cohorts, or time-sensitive decision-making problems [6, 17]. To

reduce the computational burden of pulsatile 3D simulations, several simplified or accelerated strategies have been proposed. The most widely used approximation is rigid-wall CFD, which substantially lowers computational cost by neglecting vascular deformation, but may misrepresent pressure–flow relations and phase-dependent stress transmission in compliant vessels [5, 6]. One-way or weakly coupled approaches further improve efficiency by transferring fluid loads to the structural domain without fully resolving feedback from wall deformation to the flow field [18]. Although effective in reducing cost, such approximations may compromise physical consistency when wall motion is non-negligible or when transient coupling effects are important. Reduced-order hemodynamic models provide another route to acceleration by replacing the full 3D flow problem with 1D or lumped parameter representations [19, 2]. These models can efficiently capture global pulsatile behavior, but they do not directly provide localized 3D vessel-wall stress distributions required for structural risk assessment. More recently, surrogate and machine-learning-based approaches have been introduced to accelerate high-fidelity cardiovascular simulations [20, 21, 22]. While these methods can substantially reduce online prediction time, they often require substantial amounts of high-fidelity training data and nontrivial offline training effort, and many operate as black-box approximations with limited interpretability. However, a practical framework that simultaneously enables localized wall stress estimation, preserves consistency with pulsatile hemodynamics, and avoids repeated high-fidelity transient simulations remains limited. Therefore, there remains a need for a computationally efficient framework that preserves the essential physics of pulsatile FSI while enabling localized 3D vessel-wall stress estimation.

Driven by this computational and clinical need, the objective of the present study is to develop an efficient snapshot-based framework that approximates localized vessel-wall stress without performing fully coupled transient 3D FSI simulations. This approach is motivated by the observation that critical biomechanical indicators are often evaluated at representative phases of the cardiac cycle, such as peak systole and end diastole [23, 24]. For example, peak-systolic wall stress is commonly examined as a representative biomechanical indicator in studies of aneurysmal vessels and pressure-loaded vascular pathologies [25, 26]. We hypothesize that estimating instantaneous structural stress states at such representative time instants provides sufficient diagnostic information, eliminating the need to resolve the full time-dependent evolution of the coupled problem.

To achieve this, the proposed framework decouples the temporal and spatial components of pulsatile hemodynamics. Specifically, it integrates a 1D reduced-order model to efficiently capture global pressure and flow-rate waveforms, from which boundary conditions are extracted at selected instants. Under these instantaneous boundary conditions, steady 3D fluid simulations are performed to reconstruct wall traction, which is then transferred to the structural domain to evaluate vessel-wall stress. By adopting this snapshot-based coupling, the present study introduces a methodology that retains a physics-based, interpretable formulation, while achieving significant computational savings compared to conventional high-fidelity transient FSI. The accuracy of this framework is evaluated against fully coupled 3D FSI simulations using an idealized coronary artery model, a subject-specific coronary artery, and a subject-specific abdominal aortic aneurysm model.

The remainder of this paper is organized as follows. Section 2 presents the governing equations, constitutive assumptions, boundary conditions, proposed snapshot-based coupling framework, geometrical models, validation strategy, and numerical implementation details. Section 3 reports the comparative results between the proposed method and the reference fully coupled FSI simulations. Section 4 discusses the implications, limitations, and applicability of the proposed framework. Finally, Section 5 summarizes the main conclusions of the study.

2. Methods

2.1. Proposed snapshot-based coupling framework

The proposed framework provides a computationally efficient alternative to fully coupled pulsatile fluid–structure interaction (FSI) responses at selected instants, while avoiding the computational cost associated with fully coupled 3D pulsatile simulations. The method consists of three sequential steps.

First, a one-dimensional (1D) pulsatile flow simulation based on a physics-based reduced-order model (ROM) is performed to compute the temporal waveforms of pressure and flow rate, denoted by $P(t)$ and $Q(t)$, throughout the vascular tree, including the inlet and outlet boundaries. From these waveforms, boundary condition values at a specific time point t_a , namely $P_{BC}(t_a)$ and $Q_{BC}(t_a)$, are extracted.

Second, the extracted boundary conditions are applied to a three-dimensional (3D) steady-state fluid simulation. Although temporally steady, this simula-

tion preserves the geometric complexity of the vascular domain and enables computation of the total wall traction, defined as the sum of pressure and viscous shear stress acting on the luminal surface.

Finally, a 3D steady structural analysis of the vessel wall is performed by prescribing the computed wall traction as a boundary condition. The resulting stress field is interpreted as an approximation of the structural response corresponding to the selected time point t_a under pulsatile physiological loading. An overview of the proposed snapshot-based coupling strategy is provided in Fig. 1.

2.1.1. One-dimensional reduced-order hemodynamics

Pulsatile blood flow is computed by solving the following one-dimensional governing equations under the assumption of a rigid vessel wall:

$$\frac{\partial Q}{\partial z} = 0, \quad (1)$$

$$\frac{\partial Q}{\partial t} + \frac{\partial}{\partial z} \left(\alpha \frac{Q^2}{S} \right) + \frac{S}{\rho} \frac{\partial P}{\partial z} = N \frac{Q}{S}, \quad (2)$$

where S denotes the cross-sectional area, z is the axial coordinate, α is the momentum correction coefficient, and N represents the viscous resistance term. A parabolic velocity profile is assumed, with $\alpha = 4/3$ and $N = 8\pi\nu$.

While these equations efficiently capture global pulsatile hemodynamics, they are limited in representing pressure losses in diseased or geometrically complex regions, such as stenoses and bifurcations. To address this limitation, a physics-based reduced-order modeling (ROM) approach developed by the authors [27] is incorporated to improve the relationship between flow rate and pressure drop.

In this approach, the pressure drop is modeled as a nonlinear function of the volumetric flow rate:

$$\Delta P = aQ + bQ^2 + c \frac{dQ}{dt}, \quad (3)$$

where Q denotes the volumetric flow rate. The coefficients a , b , and c represent viscous, nonlinear, and inertial contributions, respectively, and are identified using a geometry-specific preprocessing procedure described in detail in the authors' previous study [27]. The resulting ROM provides rapid computation of pressure and flow waveforms while retaining geometry-specific hemodynamic characteristics.

2.1.2. Snapshot selection and steady fluid–structure coupling

Using the reduced-order model, pressure and flow rate waveforms can be efficiently obtained at arbitrary spatial locations over the cardiac cycle. From these results, a set of representative time instants t_a is selected to capture key phases associated with significant structural response.

For each selected time point, a steady three-dimensional fluid simulation is performed by prescribing the corresponding snapshot boundary conditions extracted from the ROM. Although temporally steady, this formulation preserves the geometric fidelity of the domain and enables efficient evaluation of wall loading. The resulting wall traction is defined as:

$$\mathbf{t}_{\text{wall}}(t_a) = \boldsymbol{\sigma}_f(t_a)\mathbf{n} \quad \text{on } \Gamma_{\text{wall}}. \quad (4)$$

The computed wall traction is then transferred to the corresponding surface of the solid domain and applied as a boundary condition for structural analysis. This transfer is performed using a surface mapping procedure to ensure consistency between the fluid and solid discretizations. The structural problem is solved under a quasi-static assumption, with fixed inlet and outlet boundaries, leading to:

$$\nabla \cdot \boldsymbol{\sigma}_s + \mathbf{f}_s = 0 \quad \text{in } \Omega_s. \quad (5)$$

This framework assumes that the instantaneous fluid-induced traction dominates the structural response at each selected time point, while transient fluid inertia has a secondary effect on stress evaluation. Accordingly, the method is designed to approximate instantaneous stress states rather than the full transient dynamics of the coupled system.

2.1.3. Three-dimensional governing equations at selected time instants

Three-dimensional fluid and structural analyses were performed by solving the incompressible Navier–Stokes equations and continuity equation in the fluid domain, and Cauchy’s equation of motion in the solid domain.

For the reference fully coupled FSI simulations, the fluid and solid domains are coupled using an arbitrary Lagrangian–Eulerian (ALE) formulation, which enables consistent treatment of mesh deformation and fluid–structure interaction at the interface.

The vessel wall is modeled as an isotropic linear elastic material. Although vascular tissues exhibit nonlinear, anisotropic, and hyperelastic behavior [28], linear elasticity has been widely adopted in computational biomechanics due

to its numerical robustness and reduced parametric uncertainty [5, 29, 30, 31]. In the present study, this simplified constitutive model is employed to isolate and evaluate the performance of the proposed computational framework, rather than to capture detailed material behavior.

2.2. Validation strategy and error metrics

The proposed framework was validated by comparing von Mises stress distributions in the solid domain against those obtained from fully coupled reference FSI simulations. Comparisons were performed at selected time instants within the cardiac cycle representing both characteristic hemodynamic phases and periods of pronounced wall motion.

The selected time instants included end-diastole, peak systole, and mid-diastole, identified from the pressure waveform. In addition, time points corresponding to the maximum and minimum rates of wall-motion-induced volume change were included, as these phases are expected to yield the largest discrepancies between the snapshot-based approximation and the reference FSI solution.

Let $\sigma_{\text{vm},i}^{\text{FSI}}(t_a)$ and $\sigma_{\text{vm},i}^{\text{Prop}}(t_a)$ denote the von Mises stress at node i and time point t_a obtained from the reference FSI simulation and the proposed method, respectively. All comparisons were conducted on an identical solid mesh to ensure node-wise consistency.

To quantify the discrepancy, a relative L_2 -norm error was computed at each selected time point:

$$e_{L_2}(t_a) = \frac{\left(\sum_{i=1}^N \left[\sigma_{\text{vm},i}^{\text{Prop}}(t_a) - \sigma_{\text{vm},i}^{\text{FSI}}(t_a) \right]^2 \right)^{1/2}}{\left(\sum_{i=1}^N \left[\sigma_{\text{vm},i}^{\text{FSI}}(t_a) \right]^2 \right)^{1/2}}, \quad (6)$$

where N is the total number of solid nodes.

The error was normalized by the L_2 -norm of the reference FSI solution at the same time point, rather than by local nodal values. This choice avoids artificially inflated relative errors during low-stress phases (e.g., diastole), where the absolute stress magnitude is inherently small. In addition to the global relative L_2 -norm error of von Mises stress, nodal absolute error distributions of von Mises stress and equivalent strain were examined to assess the spatial localization of the discrepancy.

2.3. Geometrical models

Three representative geometrical models were considered to evaluate the proposed framework across increasing levels of anatomical and hemodynamic complexity. The geometrical configurations, boundary-condition assignments, and case-specific parameters are shown in Fig. 2–4.

An idealized model was constructed to represent a simplified segment of the left anterior descending (LAD) coronary artery, as shown in Fig. 2. The model has a total length of $L = 10$ cm, a lumen radius of $R = 0.1$ cm, and a wall thickness of $t = 0.02$ cm. To mimic stenotic disease, an eccentric stenosis with a degree of stenosis (DOS) of 55% was introduced in the distal segment.

A subject-specific coronary artery geometry was reconstructed from computed tomography (CT) images to assess the framework under anatomically realistic conditions, as shown in Fig. 3. Three stenotic lesions were identified in the left main (LM), mid-LAD, and distal LAD segments, with an average DOS of 69%.

A subject-specific abdominal aortic aneurysm (AAA) geometry was reconstructed from CT images to evaluate the applicability of the framework to vessels exhibiting significant luminal enlargement and complex flow patterns, as shown in Fig. 4. This case was included to assess the robustness of the proposed method in the presence of substantial geometric expansion and associated hemodynamic complexity.

The fluid and solid domains were constructed using tetrahedral meshes generated with the commercial mesh generation software MeshSim package (Simmetrix Inc., Clifton Park, NY, USA). For the fluid domain, three boundary layers were additionally generated near the wall to improve resolution of near-wall hemodynamics.

2.4. Computational details

All 3D simulations, including the steady fluid analyses, prestressing analyses, pulsatile rigid-wall simulations, and fully coupled FSI simulations, were performed using `svFSI` [32]. These simulations were performed on a high-performance computing cluster equipped with Intel[™] Xeon processors (96 cores across two sockets) and 257 GB of RAM. The 1D ROM simulations were conducted using an in-house C++ solver on a desktop system equipped with an Intel[™] Core i7-12700K 3.61 GHz processor and 64 GB of RAM.

To obtain the prestressed configuration of the vessel wall, cycle-averaged inlet pressure and outlet flow rates were computed from the converged pulsatile solution and applied in a steady-state fluid simulation. The resulting

time-averaged wall traction was subsequently imposed on the structural domain in a prestressing analysis. This prestressed configuration was used to initialize all structural simulations, including both the fully coupled FSI simulations and the proposed snapshot-based framework.

For validation, additional pulsatile rigid-wall 3D fluid simulations were performed until convergence to compare reduced-order hemodynamic predictions against high-fidelity 3D fluid solutions in the absence of wall deformation.

2.4.1. Boundary conditions and material properties

Hemodynamic boundary conditions were prescribed according to the characteristics of each geometrical model. The case-specific boundary condition configurations and parameter values are shown together with the corresponding geometries in Fig. 2–4.

For the solid domain, the inlet and outlet cross-sections were fixed in all cases, and fluid-induced traction was applied to the luminal surface. This boundary condition framework was consistently employed in both the fully coupled reference FSI simulations and the proposed snapshot-based method to ensure direct comparability.

Representative material properties were assigned based on literature values. For coronary arteries, the vessel wall was modeled with a Young’s modulus of $E = 1.0 \times 10^7$ dynes/cm², Poisson’s ratio $\nu = 0.45$, and density $\rho_s = 1.1$ g/cm³ [13, 33, 34, 35]. For the abdominal aortic aneurysm (AAA) wall, a stiffer response was assumed with $E = 6.0 \times 10^7$ dynes/cm², $\nu = 0.49$, and $\rho_s = 1.0$ g/cm³ [36, 37].

Under the quasi-static assumption adopted in the structural analysis, the inertial term ($\rho_s \frac{\partial^2 \mathbf{d}_s}{\partial t^2}$) was neglected. Blood was modeled as an incompressible Newtonian fluid with density $\rho_f = 1.06$ g/cm³ and dynamic viscosity $\mu_f = 0.04$ poise.

3. Results

This section evaluates the proposed snapshot-based framework using three representative models: an idealized left anterior descending (LAD) coronary artery model, a subject-specific coronary artery model, and a subject-specific abdominal aortic aneurysm (AAA) model. For each case, hemodynamic responses obtained from the one-dimensional (1D) reduced-order model, the

pulsatile rigid-wall three-dimensional (3D) fluid simulation, and the fully coupled FSI simulation were compared.

Across all models, the 1D reduced-order and rigid-wall 3D simulations showed close agreement in the predicted pressure and flow waveforms, indicating that the reduced-order model accurately captures global hemodynamic behavior. In contrast, discrepancies were observed when compared with the fully coupled FSI solution, highlighting the influence of wall motion. The subsequent analysis therefore focuses on quantifying the impact of wall deformation on hemodynamic responses and on the resulting stress errors of the proposed framework.

Five representative time points were defined over the cardiac cycle: end-diastole (t_1), maximum wall-induced volume-change rate (t_2), peak systole (t_3), minimum wall-induced volume-change rate (t_4), and mid-diastole (t_5). In the subject-specific coronary artery and AAA models, t_2 and t_4 occurred near the acceleration and deceleration phases of the flow-rate waveform, respectively. In the idealized LAD model, an additional acceleration-phase instant (t^*) was included because the acceleration phase did not coincide with the extrema of the wall-induced volume-change rate.

3.1. Idealized LAD coronary artery model

The hemodynamic responses of the idealized LAD model are shown in Fig. 5. The selected representative time points, wall-induced volume-change rate, pressure waveforms, and flow-rate waveforms are shown in Fig. 5(a)–(d), respectively. Since the inlet pressure waveform was prescribed, all three approaches yielded identical inlet pressure profiles, and close agreement was also observed at the outlet.

The primary differences appeared in the flow-rate response of the fully coupled FSI simulation. Positive wall-induced volume-change rates during early systole, whereas negative values appeared during late systole, corresponding to vessel expansion followed by contraction. During the expansion phase, the inlet flow rate in the FSI solution exceeded that of the rigid-wall simulation, whereas the opposite trend was observed during contraction. The outlet flow-rate waveform showed a consistent temporal pattern.

The largest discrepancies occurred near the time points corresponding to the extrema of the wall-induced volume-change rate. These points, together with the additional acceleration-phase point t^* , are indicated in Fig. 5(a). The global relative L_2 -norm errors of the von Mises stress field are summarized in Table 1. Across all evaluated time instants, the error remained below 1%.

The smallest errors were observed at end-diastole and mid-diastole, whereas relatively larger errors occurred during phases of rapid wall motion and at the additional acceleration-phase instant.

3.2. Subject-specific coronary artery model

The hemodynamic responses of the subject-specific coronary artery model are shown in Fig. 6. The pressure waveforms in the right coronary branches closely followed the inlet waveform, whereas branches distal to the left main (LM) stenosis showed pronounced distortion and temporal delay. This effect was most evident in the left coronary tree, reflecting the presence of serial stenotic segments.

At the inlet, the flow-rate waveform was nearly identical among the reduced-order, rigid-wall, and fully coupled simulations, as it was governed by the same heart model. In contrast, branch-level differences emerged at the outlets. The fully coupled FSI solution exhibited significant wall-induced volume-change rates during systole, corresponding to vessel expansion followed by contraction. During expansion, outlet flow rates in the LAD and left circumflex (LCX) branches were lower than those predicted by the rigid-wall simulation, whereas the opposite trend was observed during contraction. In comparison, the right coronary artery (RCA) branches showed only minor deviations.

The representative time points are indicated in Fig. 6(a), and the corresponding global relative errors of the von Mises stress field are summarized in Table 1. Across all evaluated time points, the error remained below 5%. Larger errors were observed during systole, particularly at time points associated with the extrema of the wall-induced volume-change rate and at peak systole.

3.3. Subject-specific aortic aneurysm model

The hemodynamic responses of the subject-specific AAA model are shown in Fig. 7. Because the inlet flow-rate waveform was prescribed, all three simulations produced identical inlet flow profiles. Consequently, differences were primarily observed in the pressure distribution and outlet flow response.

The wall-induced volume-change rate again exhibited positive values during early systole and negative values during late systole, indicating expansion followed by contraction of the aneurysmal wall. During expansion, the FSI solution predicted reduced outlet flow rates relative to the rigid-wall simulation, whereas increased flow rates were observed during contraction. In

addition, pressure differences emerged during systole: the fully coupled solution yielded higher inlet pressure and lower outlet pressure than the rigid-wall case, resulting in a larger pressure drop across the aneurysmal segment.

The selected time instants are marked in Fig. 7(a), and the corresponding global relative L_2 -norm errors of the von Mises stress field are reported in Table 1. The error remained below 2% at all evaluated instants. As in the coronary artery model, relatively larger errors were observed during systole, particularly near the extrema of the volume-change rate and at peak systole.

3.4. Computational cost comparison

The computational cost of the proposed framework and the reference fully coupled FSI simulations is summarized in Table 2. For the proposed framework, the total cost comprises four components: (i) a transient ramped-flow simulation for identification of the ROM coefficients, (ii) a 1D pulsatile simulation, (iii) steady 3D fluid simulations, and (iv) steady structural simulations performed at the selected time points. The transient ramped-flow simulation is required only once for a given geometry; once the ROM coefficients are identified, additional snapshot evaluations can be performed without repeating this preprocessing step.

In contrast, the reference FSI analysis requires time-resolved 3D fluid–structure coupling over multiple cardiac cycles until a periodic solution is achieved. The primary source of computational savings in the proposed framework therefore lies in replacing the fully pulsatile 3D simulations with reduced-order hemodynamics and steady 3D snapshot evaluations.

Across all three models, the proposed framework reduced the overall computational cost by more than an order of magnitude, exceeding a 30-fold reduction in all cases. The reduction was more pronounced for geometries requiring finer spatial resolution and longer simulation times to reach periodicity in the fully coupled FSI analysis. Detailed timings for each simulation component are reported in Table 2.

4. Discussion

This study presents a snapshot-based computational framework for efficient estimation of localized vessel-wall stress under pulsatile hemodynamic loading. By combining a one-dimensional (1D) reduced-order model (ROM) for rapid evaluation of time-dependent boundary conditions with steady three-dimensional (3D) fluid and quasi-static structural analyses at selected

time instants, the proposed approach avoids repeated fully coupled transient 3D FSI simulations while preserving the spatial resolution required for localized stress evaluation.

Across all three models, the proposed framework reproduced the vessel-wall stress obtained from the reference FSI simulations with low global relative L_2 -norm error, while achieving a substantial reduction in computational cost. The error remained small over the selected cardiac phases and increased primarily during periods of rapid wall motion. These results indicate that the framework captures the dominant features of the instantaneous structural stress state without resolving the full transient FSI problem.

A discrepancy between the proposed framework and the reference FSI solution is expected, as the transient coupled problem is replaced by a sequence of time-specific approximations. In the present approach, the structural stress field at each time point is determined from wall traction obtained from a steady rigid-wall fluid simulation driven by ROM-derived boundary conditions. Since the 1D ROM and the pulsatile rigid-wall 3D simulations showed close agreement in the hemodynamic waveforms, the dominant discrepancy is unlikely to arise from the reduced-order hemodynamic prediction itself. Rather, it is more likely associated with the reconstruction of instantaneous wall traction when transient fluid inertia, wall-motion feedback, and deformation-induced changes in the local pressure–flow relation are not fully resolved.

The role of wall motion can be interpreted in terms of a temporary storage effect. During wall expansion, part of the inflow may be accommodated by the increase in luminal volume rather than being transmitted immediately downstream. During recoil, the stored volume is released back into the flow field. This phase-dependent storage and release can introduce a small temporal lag between the rigid-wall and compliant-wall responses. Although this lag may be modest in the waveform comparison, it can produce appreciable instantaneous differences at selected time points, particularly near phases of rapid wall motion. These differences modify the local pressure field and reconstructed wall traction, which are then reflected in the nodal stress field.

The magnitude and spatial distribution of this traction mismatch depend on the vascular configuration and boundary conditions. In the idealized coronary artery model, where the inlet pressure waveform was prescribed, wall expansion in the FSI solution induced additional inflow, whereas recoil reduced it. In the subject-specific coronary and aneurysm models, the inlet flow was constrained by the heart model or by the prescribed inflow waveform. In

these cases, wall motion primarily altered downstream flow transmission and branch-level flow distribution rather than the imposed inflow itself. In the coronary model, this effect was more evident in the left coronary branches, where serial stenoses increased hydraulic resistance and likely enhanced delayed downstream transmission. The right coronary branches showed smaller deviations, consistent with a less restricted flow pathway. In the aneurysm model, expansion of the aneurysmal sac may similarly have acted as a temporary compliant reservoir, delaying transmission of flow toward the iliac outlets. These mechanisms provide a plausible explanation for the larger errors observed near the selected expansion and recoil phases.

Geometry-dependent flow features and local structural response further modulated how hemodynamic discrepancies were translated into spatial error. As shown in Fig. 8, the absolute error in von Mises stress was small over most of the vessel wall in all three models, despite being evaluated at instantaneous time points where phase-dependent discrepancies were expected to be most pronounced. In the idealized coronary model, the stress error remained localized near the stenotic segment, with a peak absolute error of 2.57×10^4 dynes/cm², compared with a stress range of approximately 5.2×10^3 – 1.32×10^6 dynes/cm². The subject-specific coronary model showed a larger local peak error of 1.50×10^5 dynes/cm², primarily near stenotic and bifurcation regions, while the remaining regions showed substantially smaller errors. In contrast, the abdominal aortic aneurysm model exhibited a more spatially distributed stress-error pattern over the aneurysmal wall, with a peak absolute error of 6.69×10^4 dynes/cm² relative to a broader stress range of 3.09×10^4 – 5.66×10^6 dynes/cm². These trends suggest that abrupt geometric transitions, such as stenoses and bifurcations, can amplify local sensitivity to traction mismatch, whereas aneurysmal expansion tends to distribute the discrepancy over a broader wall region.

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A similar tendency was observed in the equivalent strain field, as shown in Fig. 9. In the idealized and subject-specific coronary models, the absolute strain error was mostly small but became locally elevated near geometrically sensitive regions. The peak absolute strain errors were 4.0×10^{-3} and 3.6×10^{-2} in the idealized and subject-specific coronary models, respectively. By contrast, the abdominal aortic aneurysm model showed a much smaller peak absolute strain error of 1.3×10^{-5} , with a relatively diffuse distribution over the aneurysmal wall. The consistency between the stress and strain error patterns indicates that the discrepancy was not confined to a single stress measure, but was also reflected in the deformation-related structural response. However, because the aneurysm model used different material properties from the coronary models, the differences among cases should be interpreted as the combined effect of geometry-dependent hemodynamics, wall compliance, and local structural response, rather than as an isolated effect of vascular morphology alone.

The temporal distribution of the error further supports this interpretation. In all models, larger errors occurred near the extrema of the wall-induced volume-change rate, corresponding to phases of rapid expansion or contraction. These are precisely the conditions under which the rigid-wall approximation deviates most from the compliant FSI response. The additional acceleration-phase instant in the idealized model followed the same trend, indicating that transient flow variation can also contribute when it is not aligned with peak wall motion. Therefore, the increased error at these time points should be interpreted primarily as a phase-dependent traction reconstruction error, rather than as a failure of the reduced-order hemodynamic model.

The magnitude of wall deformation observed in the reference FSI simulations was physiologically reasonable. The coronary models exhibited diameter variations of approximately 3%–5% over the cardiac cycle, consistent with reported *in vivo* measurements [38], whereas the aneurysm model showed

smaller variations of approximately 1%, in agreement with clinical observations [39].

Although the present study focused on methodological validation, the proposed framework has potential applications in time-specific structural assessment of vascular systems. Biomechanical metrics such as peak systolic stress or stress variation across the cardiac cycle are commonly used to identify mechanically vulnerable regions. Because the proposed approach reproduces stress fields at representative time instants with low global error, it provides a computationally efficient alternative for repeated patient-specific evaluations where fully coupled FSI simulations are impractical.

Several limitations of this study should be acknowledged. First, the accuracy of the proposed framework depends on the fidelity of the reduced-order model (ROM), and any error in the reduced-order hemodynamic predictions propagates through the subsequent analyses. Second, the steady rigid-wall formulation used in the 3D fluid step neglects transient flow effects and fluid–structure feedback, including the phase-dependent influence of wall deformation on local pressure and flow redistribution. As a result, discrepancies may arise during phases of rapid wall motion. Third, the vessel wall was modeled as a linear elastic and homogeneous material, which does not capture the nonlinear, anisotropic, and heterogeneous properties of vascular tissue *in vivo*. Fourth, plaque composition was not explicitly modeled in the coronary cases, limiting the interpretation of local stress magnitudes in regions with heterogeneous lesion characteristics. Finally, validation was performed at selected time instants rather than over the full transient response. Accordingly, the proposed framework should be interpreted as an efficient approximation of instantaneous stress states, rather than a surrogate for fully time-resolved FSI simulations.

Despite these limitations, the proposed framework preserves the dominant features of localized vessel-wall stress while substantially reducing computational cost. This balance between efficiency and physical fidelity makes it a promising tool for applications in which full transient FSI analysis is computationally prohibitive but time-specific structural information remains essential.

5. Conclusion

This study presented a snapshot-based computational framework for efficient estimation of localized vessel-wall stress under pulsatile hemodynamic

loading. The method combines a 1D reduced-order model for rapid evaluation of time-dependent boundary conditions with steady 3D fluid and quasi-static solid analyses at selected time instants. In this way, the proposed approach avoids repeated fully coupled transient 3D FSI simulation while retaining the spatial resolution required for localized stress evaluation.

The framework was assessed in an idealized coronary artery model, a subject-specific coronary artery model, and a subject-specific aneurysm model. In all three cases, the proposed method reproduced the von Mises stress obtained from the reference 3D FSI simulations with low global relative L_2 -norm error at representative time instants. The larger discrepancies were observed near phases of rapid wall motion, whereas smaller errors were found at end-diastole and mid-diastole. These results indicate that the proposed framework can capture the dominant instantaneous stress state with substantially lower computational cost than full pulsatile 3D FSI analysis.

The remaining error was primarily associated with differences in the reconstructed fluid loading, especially during phases at which wall motion altered the effective pressure–flow relation and downstream flow transmission. Nevertheless, the overall agreement across vascular models with different geometric and hemodynamic characteristics supports the applicability of the proposed strategy for time-specific structural assessment. The present framework may therefore provide a practical basis for computationally efficient, time-specific structural assessment in cardiovascular biomechanics, particularly in applications for which repeated full 3D FSI simulation is not feasible.

Time point	Idealized coronary artery model		Subject-specific coronary model		Subject-specific aneurysm model	
	Time (s)	Error (%)	Time (s)	Error (%)	Time (s)	Error (%)
t_1 : End-diastole	0.05	0.13	0.05	1.98	0.08	0.22
t_2 : Max. vol.-change rate	0.15	0.96	0.15	3.04	0.15	1.12
t_3 : Peak systole	0.28	0.28	0.25	3.91	0.25	1.00
t_4 : Min. vol.-change rate	0.35	0.91	0.35	4.45	0.35	0.87
t_5 : Mid-diastole	0.80	0.15	0.70	1.62	0.70	0.30
t^* : Additional phase	0.45	0.17	—	—	—	—

Table 1: Selected time instants and corresponding global relative L_2 -norm error of the von Mises stress field for the three models. Here, vol.-change rate denotes the wall-motion-induced volume-change rate.

Simulation stage		Idealized coronary artery model	Subject-specific coronary model	Subject-specific aneurysm model
Proposed framework	Ramped-flow ^a	3.3	33	5
	1D pulsatile	0.2	3	0.9
	3D steady fluid	1.2	15	7
	3D steady solid	8	65	21
	Total	12	116	34
Full 3D FSI		6866	9989	1030

All times are reported in minutes.

^a One-time preprocessing step for a given geometry; not repeated for additional snapshot evaluations.

Table 2: Computational time required for the proposed snapshot-based framework and the reference 3D FSI simulation in the three models.

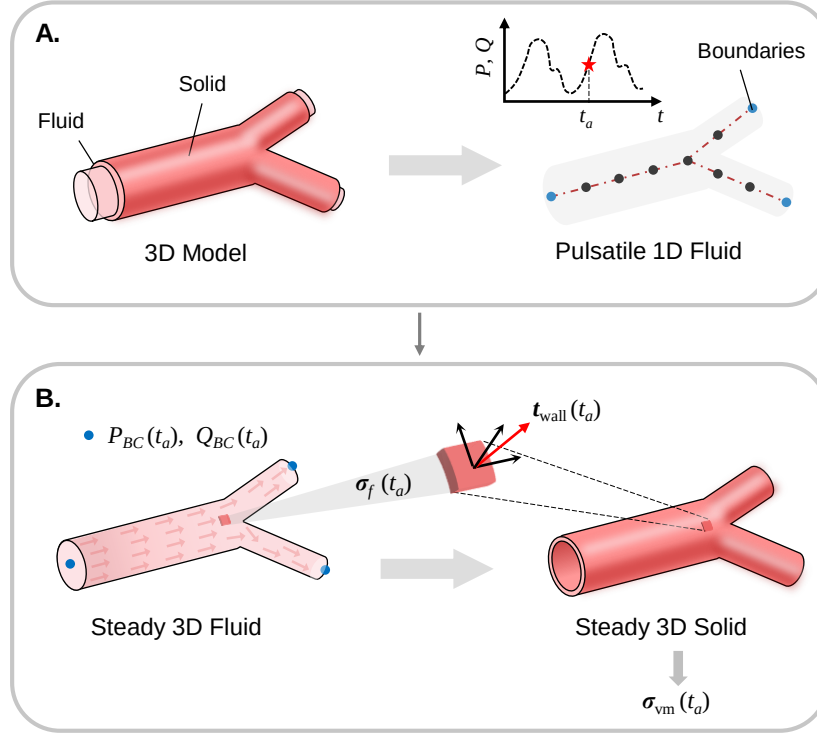


Figure 1: Overview of the proposed snapshot-based coupling framework. (A) The 3D vascular geometry is reduced to a 1D representation to compute pulsatile hemodynamics. (B) At selected time instants, the extracted snapshot boundary conditions are applied to steady 3D fluid and structural analyses to estimate localized vessel-wall von Mises stress.

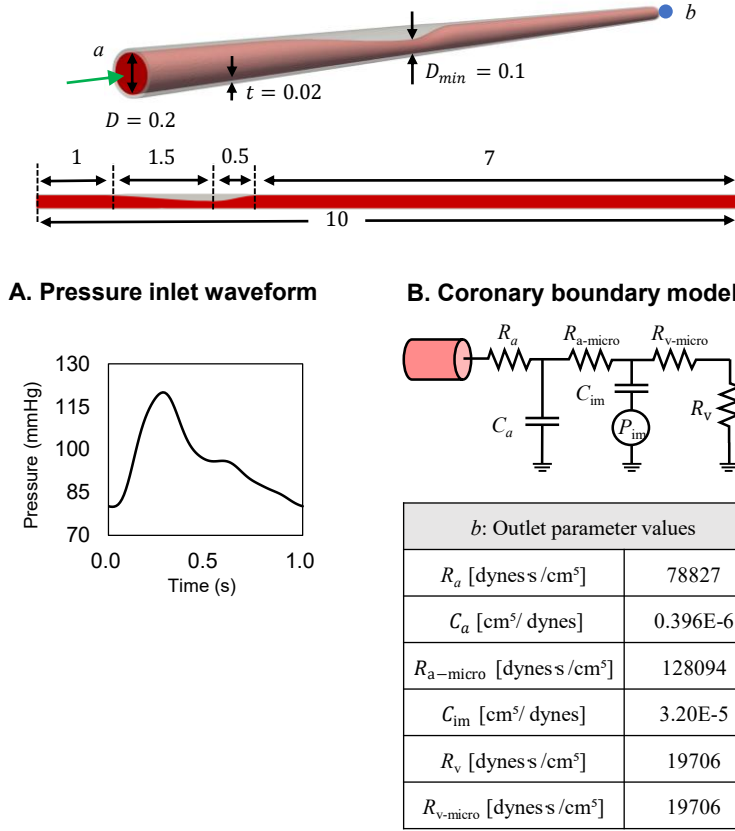
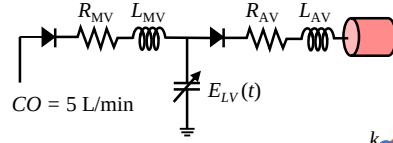
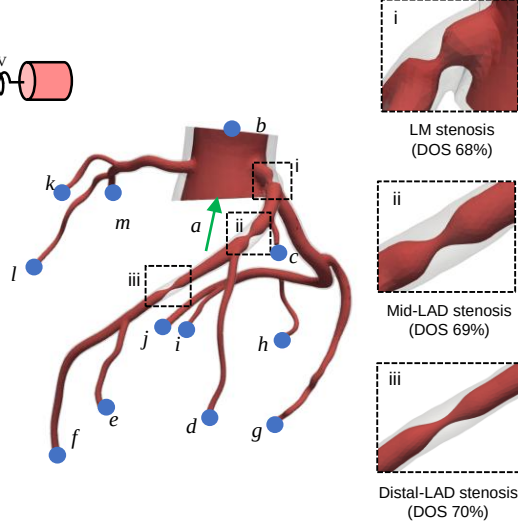


Figure 2: Idealized coronary artery model used in this study. The figure shows the simplified cylindrical geometry with dimensional information and an eccentric stenosis. The green arrow indicates the inlet location, and the blue dot indicates the outlet location. (A) Prescribed inlet pressure waveform. (B) Coronary lumped-parameter boundary model and the corresponding parameter values.

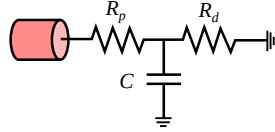
A. Heart boundary model



a: Inlet parameter values	
CO [L/min]	5.0
t_{cycle} [s]	1.0
t_{sys} [s]	0.33
R_{LA} [dyn·s/cm ⁵]	5.0
L_{LA} [dyn·s ² /cm ⁵]	1.0
R_{LV} [dyn·s/cm ⁵]	10.0
L_{LV} [dyn·s ² /cm ⁵]	10.0
E_{max} [mmHg/cc]	2.0

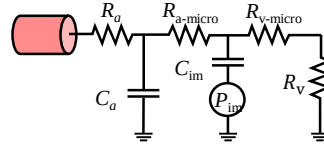


B. Windkessel RCR



b: Outlet parameter values	
R_p [dyn·s/cm ⁵]	232.44
C [cm ⁵ /dynes]	1.50E-3
R_d [dyn·s/cm ⁵]	1317.18

C. Coronary boundary model



c-l: Outlet parameter values					
Outlets	R_a	$R_{a-micro}$	$R_{v-micro}$	C_a	C_{im}
c	76680	128624	39576	2.86E-7	2.31E-6
d	67588	113374	34884	3.97E-7	3.21E-6
e	65168	109315	33635	4.37E-7	3.53E-6
f	56556	94868	29190	6.31E-7	5.11E-6
g	64918	108895	33506	4.41E-7	3.57E-6
h	70224	117796	36244	3.60E-7	2.91E-6
i	71302	119603	36801	3.45E-7	2.79E-6
j	46302	77668	23898	1.06E-6	8.60E-6
k	66831	112104	34493	7.50E-7	6.07E-6
l	65058	109130	33578	8.04E-7	6.51E-6
m	55847	93680	28824	1.20E-7	9.70E-7

* R [dyn·s/cm⁵], C [cm⁵/dynes]

Figure 3: Subject-specific coronary artery model used in this study. The figure shows the reconstructed coronary geometry with serial triple stenoses. The green arrow indicates the inlet location, and blue dots indicate outlet locations. (A) Heart boundary model. (B) Windkessel RCR boundary model. (C) Coronary lumped-parameter boundary model. Each panel includes the corresponding lumped-parameter circuit and parameter values.

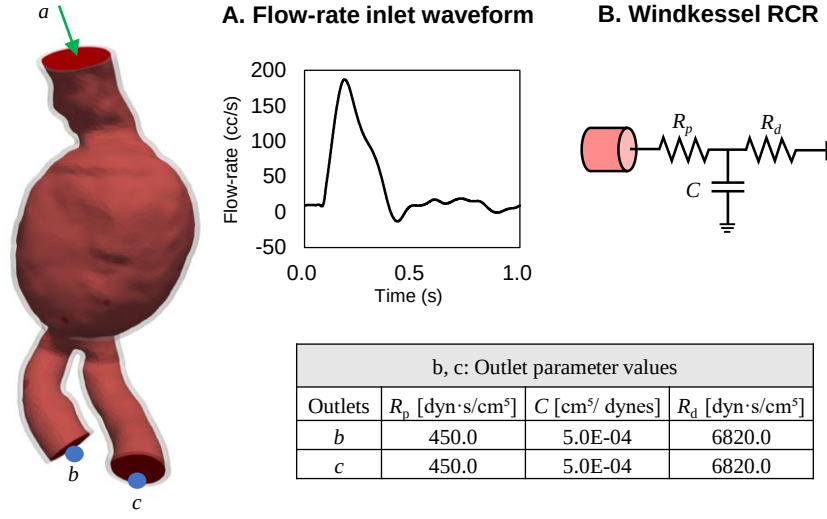


Figure 4: Subject-specific abdominal aortic aneurysm (AAA) model used in this study. The figure shows the reconstructed aneurysmal aortic geometry. The green arrow indicates the inlet location, and blue dots indicate outlet locations. (A) Prescribed inlet flow-rate waveform. (B) Windkessel RCR boundary model and the corresponding parameter values.

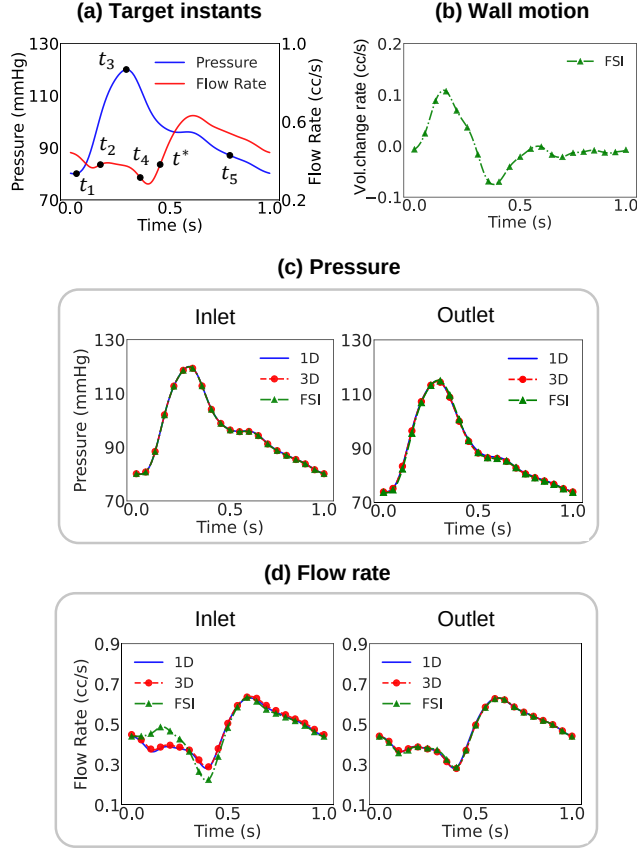


Figure 5: Comparison of hemodynamic responses in the idealized coronary artery model obtained from the pulsatile 1D reduced-order simulation, the rigid-wall pulsatile 3D simulation, and the fully coupled pulsatile FSI simulation. (a) Selected representative time points. (b) Wall-motion-induced volume-change rate. (c) Pressure waveforms. (d) Flow-rate waveforms. An additional acceleration-phase instant, (t^*), is included for this model.

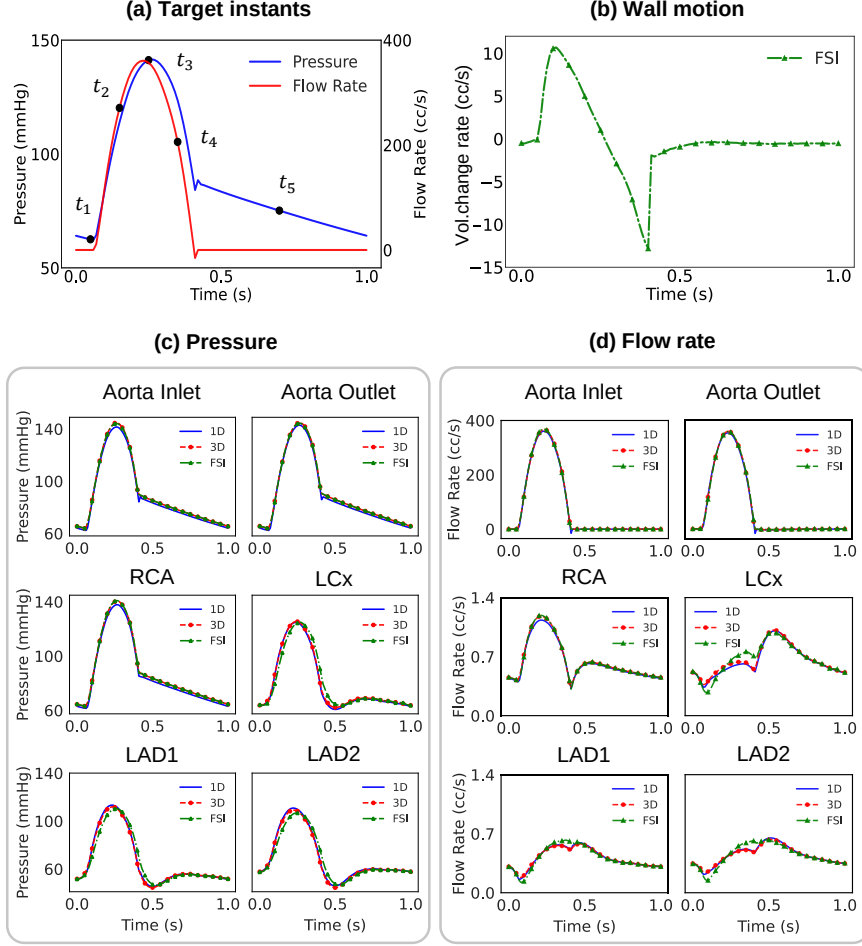


Figure 6: Comparison of hemodynamic responses in the subject-specific coronary artery model obtained from the pulsatile 1D reduced-order simulation, the rigid-wall pulsatile 3D simulation, and the fully coupled pulsatile FSI simulation. Representative inlet and outlet responses are shown for the aortic inlet and selected coronary branches. LAD1 denotes the diagonal branch distal to a stenotic lesion, and LAD2 denotes the distal LAD segment downstream of all stenotic lesions. (a) Selected representative time points. (b) Wall-motion-induced volume-change rate. (c) Pressure waveforms. (d) Flow-rate waveforms.

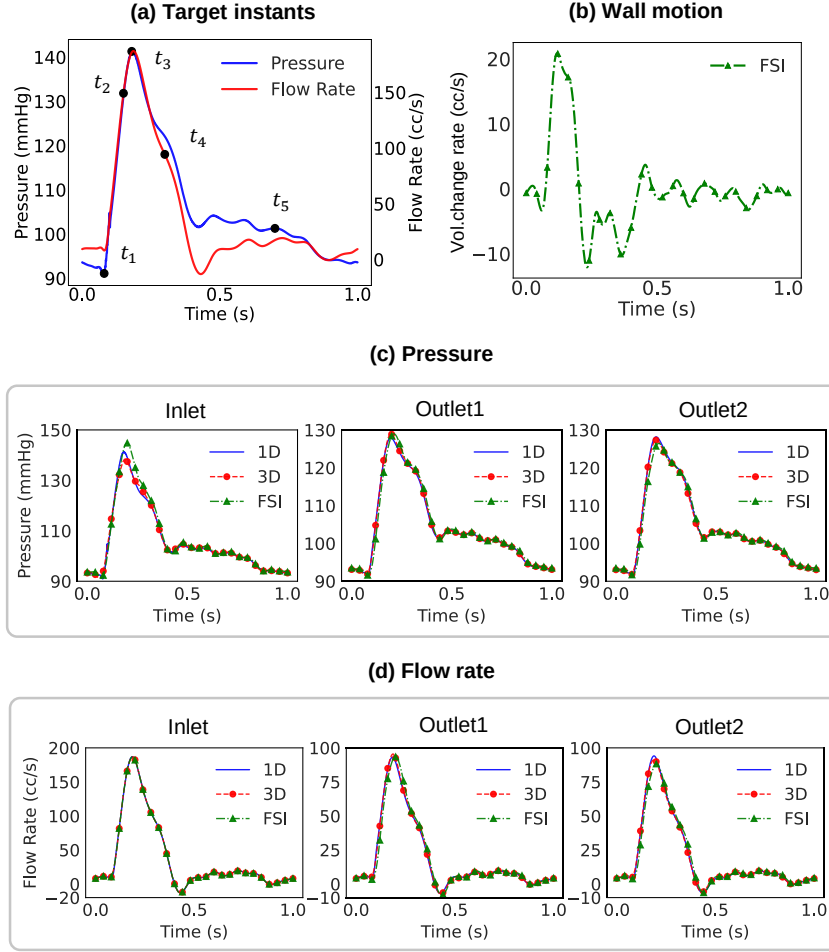


Figure 7: Comparison of hemodynamic responses in the subject-specific abdominal aortic aneurysm (AAA) model obtained from the pulsatile 1D reduced-order simulation, the rigid-wall pulsatile 3D simulation, and the fully coupled pulsatile FSI simulation. Inlet and outlet responses are shown. (a) Selected representative time points. (b) Wall-motion-induced volume-change rate. (c) Pressure waveforms. (d) Flow-rate waveforms.

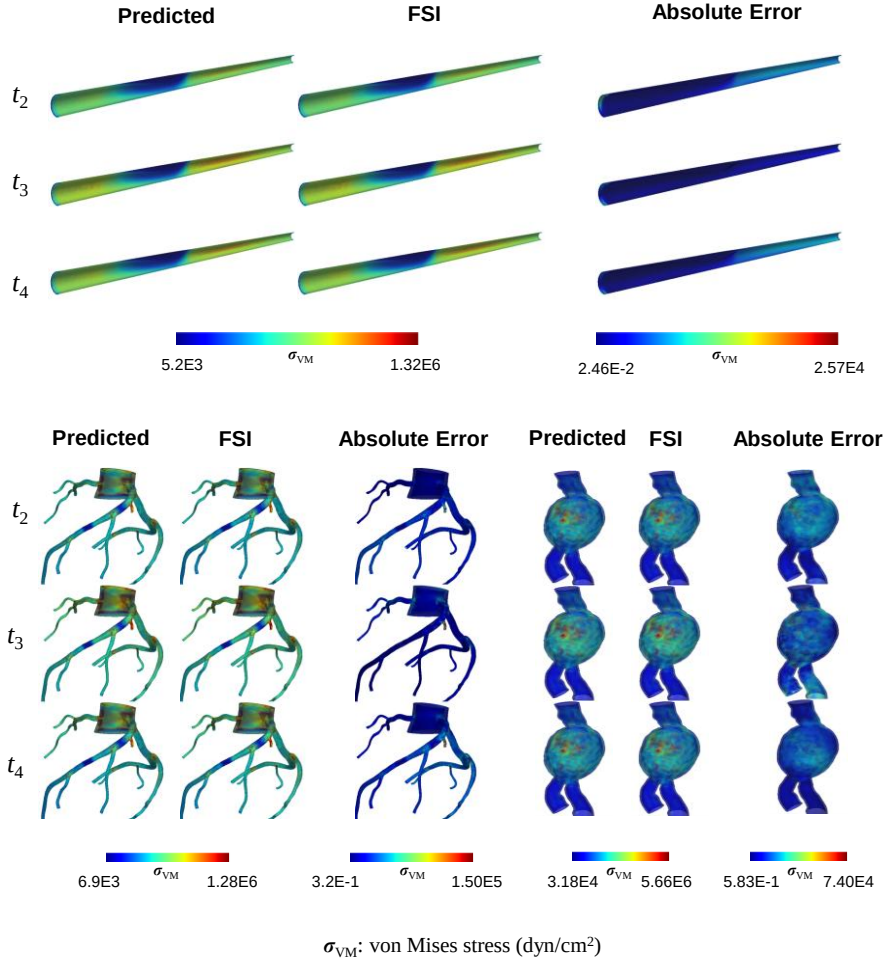


Figure 8: Spatial distributions of von Mises stress at t_2 , t_3 , and t_4 for the idealized coronary artery model, the subject-specific coronary artery model, and the subject-specific abdominal aortic aneurysm model. For each model, the predicted result from the proposed framework, the reference FSI result, and the corresponding nodal absolute error are shown.

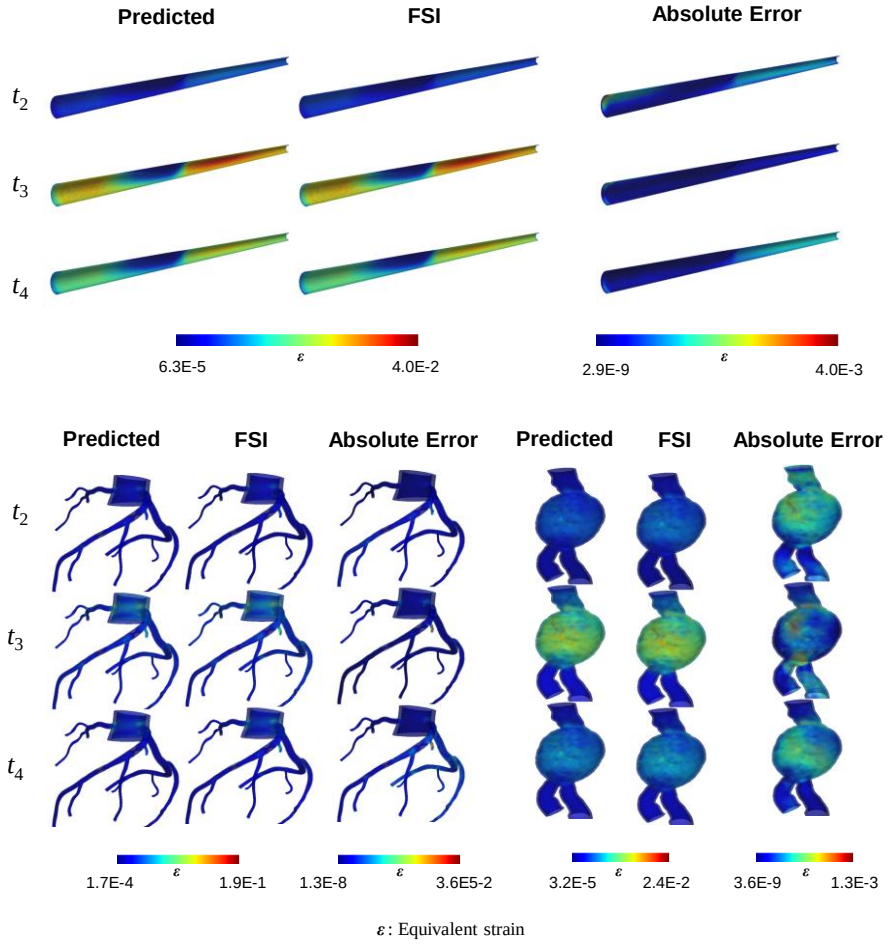


Figure 9: Spatial distributions of equivalent strain at t_2 , t_3 , and t_4 for the idealized coronary artery model, the subject-specific coronary artery model, and the subject-specific abdominal aortic aneurysm model. For each model, the predicted result from the proposed framework, the reference FSI result, and the corresponding nodal absolute error are shown.

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